NLP Formulas Cheat Sheet

1. KL Divergence

$$D(p \parallel q) = \sum_x p(x) \log_2 \frac{p(x)}{q(x)}$$

2. Jaccard Similarity

$$J(A,B) = \frac{|A \cap B|}{|A \cup B|}$$

3. Minimum Edit Distance

$$D(i,j) = \min \begin{cases} D(i-1,j)+1 \\ D(i,j-1)+1 \\ D(i-1,j-1)+\text{cost} \end{cases}$$

(cost $= 0$ if chars match, else 1)

4. Skip-gram Parameters

$$\text{Params} = 2 \times V \times d$$

5. Bigram MLE

$$P(w_i \mid w_{i-1}) = \frac{\text{Count}(w_{i-1}, w_i)}{\text{Count}(w_{i-1})}$$

6. Perplexity

$$PP(W) = \left(\prod_{i=1}^{N} \frac{1}{P(w_i \mid w_{i-1})} \right)^{\frac{1}{N}}$$

7. Add-1 Smoothing

$$P(w_i \mid w_{i-1}) = \frac{\text{Count}(w_{i-1}, w_i) + 1}{\text{Count}(w_{i-1}) + V}$$

8. HMM Sequence Probability

$$P = \prod \text{Transition} \times \prod \text{Emission}$$

9. Heaps' Law

$$V = k \cdot N^{\beta}$$

10. LDA (Gibbs Sampling)

Topic–Word:

$$\beta_w^{(k)} = \frac{n_w^{(k)} + \eta}{\sum_w n_w^{(k)} + V\eta}$$

Document–Topic:

$$\theta_k^{(d)} = \frac{n_k^{(d)} + \alpha}{\sum_k n_k^{(d)} + K\alpha}$$

11. Cosine Similarity

$$\cos\theta = \frac{\mathbf{A} \cdot \mathbf{B}}{\|\mathbf{A}\| \, \|\mathbf{B}\|}$$

12. Evaluation Metrics

$$\text{Precision} = \frac{TP}{TP + FP}, \quad \text{Recall} = \frac{TP}{TP + FN}$$

$$F1 = \frac{2 \cdot \text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

13. Zipf's Law

$$f(r) = \frac{C}{r}$$